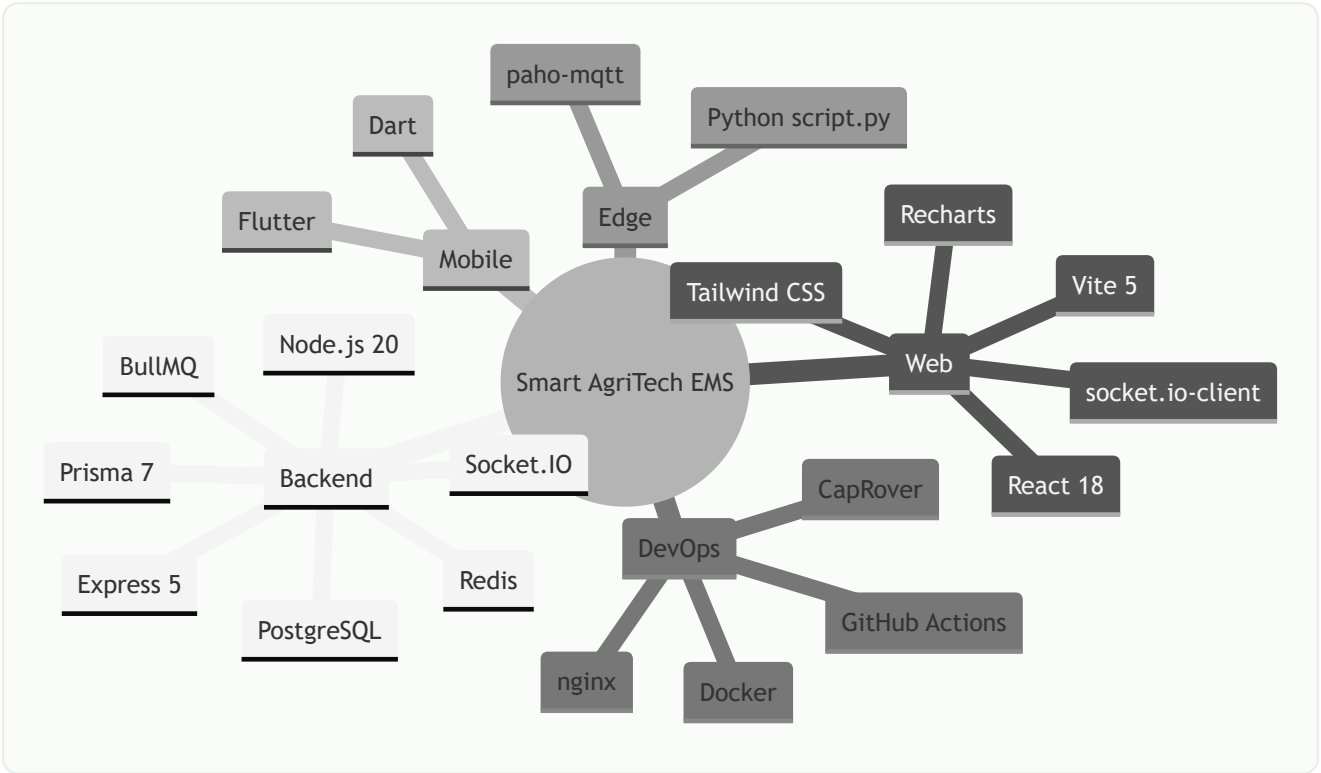


Tech stack

Technologies used across the Smart AgriTech EMS monorepo.

Stack at a glance



Backend (`ems/ems-backend`)

Layer	Technology	Version / notes
Runtime	Node.js	20 LTS (Alpine in Docker)
Framework	Express	5.x
ORM	Prisma	7.x with <code>@prisma/adapters-pg</code>
Database	PostgreSQL	16 recommended; TimescaleDB optional
Cache / queue	Redis	7.x; required for production ingest batching
Job queue	BullMQ	Ingest, anomaly, email, device-delete workers

Layer	Technology	Version / notes
Real-time	Socket.IO	+ optional <code>@socket.io/redis-adapter</code>
Auth	jsonwebtoken	Access + refresh token rotation
Validation	express + custom	Controllers + <code>AppError</code>
Email	Nodemailer	Gmail or custom SMTP / SES
Media	Cloudinary	Icons, product images (optional)
Scheduling	node-cron	In-process scheduled device tasks
Metrics	Custom Prometheus text	<code>/metrics</code> endpoint
Security	helmet, cors, compression	Production hardening
Dev tools	deviceSimulator, seedFleet	Load testing & demo data

Key files:

- `server.js` — application entry
- `prisma/schema.prisma` — data model (~40 models)
- `workers/jobQueues.js` — async processing
- `services/ingestService.js` — telemetry hot path

Web frontend (`web_frontend`)

Layer	Technology	Notes
UI library	React	18.x
Build	Vite	5.x; dev proxy to API :5000
Styling	Tailwind CSS	Custom design tokens in <code>index.css</code>
Routing	React Router	6.x; role-based route trees
Charts	Recharts	Dashboards, historical data, AI pages
HTTP	fetch API	<code>api/client.js</code> with JWT refresh
Real-time	socket.io-client	Live readings & alarms
UI snapshots	boneyard-js	Skeleton loading states
Icons	lucide-react	Sidebar and actions
Production serve	nginx:alpine	SPA + gzip; Docker multi-stage build

Build-time env:

Variable	Purpose
<code>VITE_API_URL</code>	REST base (e.g. <code>https://api.example.com/api</code>)
<code>VITE_SOCKET_URL</code>	Socket.IO origin (e.g. <code>https://api.example.com</code>)
<code>VITE_ENABLE_SOCKET</code>	Set <code>false</code> to disable WebSocket client

Mobile app (`app/`)

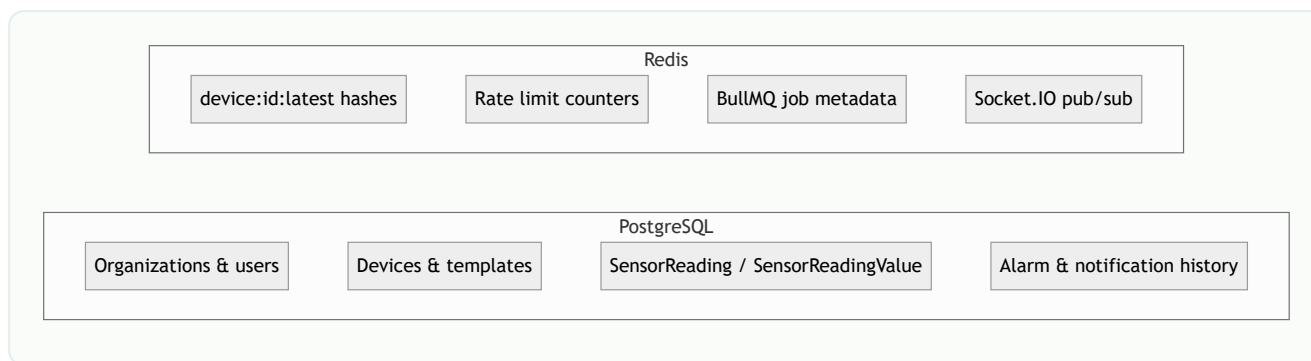
Layer	Technology
Framework	Flutter
Language	Dart
API	Same REST backend as web
Platforms	Android / iOS (standard Flutter targets)

The mobile app mirrors core EMS capabilities for field users; web dashboard is the primary admin surface.

Edge / integration

Component	Stack	Purpose
<code>script.py</code>	Python 3, paho-mqtt	Subscribe MQTT soil topic → HTTP ingest
<code>deviceSimulator.js</code>	Node.js	Synthetic fleet telemetry for testing
<code>seedFleet.js</code>	Node.js + Prisma	100 devices / 10 gateways demo fleet

Data stores



Store	What lives there
PostgreSQL	All durable entities, audit history, commands
Redis	Latest readings cache, ingest queue, optional rate limits & socket cluster

DevOps & deployment

Tool	Role
Docker	Backend & frontend container images
CapRover	PaaS on VPS — HTTPS, app deploy, one-click Postgres/Redis
GitHub Actions	<code>caprover-deploy-backend.yml</code> , <code>caprover-deploy-frontend.yml</code>
CapRover CLI	<code>caprover deploy -t tarball</code> from CI
nginx	Static frontend + gzip
Let's Encrypt	Via CapRover HTTP settings

Pack scripts:

- `scripts/caprover/pack-backend.sh`
- `scripts/caprover/pack-frontend.sh`
- `scripts/caprover/pack-all.ps1` (Windows manual)

Development tooling

Tool	Usage
npm	Backend & frontend package management

Tool	Usage
Prisma CLI	<code>migrate deploy</code> , <code>generate</code> , <code>db seed</code>
Git	Monorepo; path-filtered CI
Vite dev server	HMR on port 5173
ESLint / IDE	Per-project conventions

Recommended minimum specs

Local development (single machine)

Resource	Minimum
CPU	4 cores
RAM	8 GB
Disk	10 GB free
Services	Node 20, PostgreSQL 16, Redis 7

Production (small deployment, <100 devices, 1 msg/sec)

Resource	Recommended
VPS	2 vCPU, 4 GB RAM, 40 GB SSD
Stack	CapRover + 4 apps (frontend, backend, postgres, redis)
OS	Ubuntu 22.04 LTS

Production (medium, 100–1000 devices)

Resource	Recommended
VPS	4 vCPU, 8–16 GB RAM, 80+ GB SSD
Redis	Dedicated or managed; required
DB	Consider managed PostgreSQL or read replica (<code>DATABASE_READ_URL</code>)

Version matrix (pin in production)

Component	Pin strategy
Node	<code>20-alpine</code> in Dockerfiles
PostgreSQL	<code>16-alpine</code> (CapRover one-click)
Redis	<code>7-alpine</code>
CapRover	<code>caprover/caprover:1.14.1+</code> (Docker API 1.44+)

Related documents

- [Architecture](#)
- [Backend](#)
- [Web frontend](#)
- [Deployment guide](#)